

Requirements Sources Rough Draft – C&C

Date:

July 7th, 2025

Researcher(s):

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Objective of research:

Find sources for the requirements I created last week and discuss them in team meeting

Notes:

Requirement 1: The ROV's microcontroller shall use Python coding for all computations

Sources:

<https://docs.micropython.org/en/latest/> - This link leads to the documentation for MicroPython, a potential way of using Python on Arduino boards

Requirement 2: The ROV shall use one Arduino Pro Mini/Nano as a microcontroller

Sources:

<https://docs.arduino.cc/hardware/nano/> - Documentation for Arduino Nano if we decided to use this type of microcontroller

<https://docs.arduino.cc/retired/boards/arduino-pro-mini/> - Documentation for Arduino Pro Mini if we decided to use this type of microcontroller

Requirement 3: The ROV shall use one camera to give the pilot an understanding of where the ROV is

Sources:

https://drive.google.com/file/d/1Wda_ZJjK9ycHMS2GFzKe2RmPIOyOglU9/view?usp=sharing - This source leads to the Merknights 2024 technical sheet which shows how the team was able to use one camera to pilot their ROV

Requirement 4: The ROV shall use **XX** inches of wire to send signals between devices

Sources:

<https://www.homebuiltrovs.com/brcs/BRCSWiringManual.pdf> - *This is a written manual on how to wire underwater ROV's*